

Week 6 Report: Model Development

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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What I did this week

This week I tuned the models with 5-fold cross-validation on the training split only, so the test set stays untouched. I used F1 as the tuning score because both missed threats and false alarms matter. The table shows the chosen settings and the cross-validated F1.

Decision Tree and Random Forest reach the same cross-validated F1 of 0.996, with a very small spread across folds (standard deviation 0.001). SVM is a little lower at 0.987.

Model	Best settings	CV F1
Decision Tree	{max_depth: 10, min_samples_leaf: 5}	0.996
Random Forest	{max_depth: 20, min_samples_leaf: 1}	0.996
SVM (RBF)	{C: 10, gamma: scale}	0.987

Next week

Evaluate the tuned models once on the held-out test set and compare them statistically.